Response to Reviewers

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"Simultaneous Contact Selection and Planning for Contact-Rich Manipulation with Cascaded Optimization"

We thank all three reviewers for their constructive comments and for highlighting the strengths of our work, including that SCSP addresses a real limitation of local contact-implicit model predictive control (CIMPC) methods (**R1**) and a real gap in existing CIMPC methods (**R2**), the proposed contact selection and ranking strategy is practical and effective (**R1**, **R3**), and that the experiments are thorough and extensive (**R3**) and above average for this literature (**R2**), with strong results across simulation and real-world hardware experiments without motion capture (**R1–R3**).

Reviewer 1

Q1: The CSO-as-MIQP claim is under-specified

Section III-D states that Eq. (32) is formulated as an MIQP, that the inner loop constitutes a QP, and that the overall CSO is an efficient MIQP. This is plausible only under additional assumptions that are not made explicit.

For a fixed sampled contact point, fixed contact geometry, fixed Jacobians, quadratic objective, polyhedral friction constraints, and an explicit active-set or binary-variable encoding of

$$\hat{\lambda}_{\text{env},i} = \max(A_i \hat{\lambda}_r + b_i, 0), \tag{1}$$

the problem can be written as a region-wise QP or as an MIQP. However, the paper does not provide the MIQP variables, big-M or indicator constraints, active-set constraints, or the precise quadratic form being solved. The implementation section also states that the CSO inner loop is solved using CasADi with SNOPT, which is an SQP/NLP solver rather than a QP or MIQP solver.

The paper should either present the exact QP/MIQP formulation and solver details, or weaken the claim to finite candidate search with local continuous force optimization under an SCM approximation.

Answer: Thank you for pointing out this inconsistency. We agree that our previous characterization of CSO as an MIQP, and of its inner problem as a QP, was imprecise.

In our actual implementation, the sampled contact location is not encoded as an integer decision variable, and we do not introduce binary variables, big-M constraints, indicator constraints, or use a mixed-integer solver. Instead, the discrete contact-location variable is handled explicitly through the finite candidate set $\mathcal{I}_{\text{avail}}$. For each candidate contact location, we solve a continuous optimization problem over the robot contact forces and then compare the resulting costs across candidates.

We need to correct an omission in our previous description. In addition to constructing a closed-form contact-dynamics approximation, SCM uses $K = 3$ unrolled Gauss–Seidel sweeps over the environment contacts (Eq. (27)), initialized with zero environment forces and the free-motion object velocity. Each sweep evaluates each contact force from the object velocity already updated by the preceding contacts and sweeps and therefore introduces a sequential coupling, so the resulting problem is even farther from a QP formulation. The previous implementation solved the continuous subproblem serially with SNOPT; the current implementation replaces this with GPU-parallel CEM. Overall, CSO is a lightweight optimization problem that permits efficient parallel solution, but it does not satisfy a QP formulation.

The revised formulation therefore does not introduce binary variables, big-M constraints, indicator constraints, or a mixed-integer solve. The previous "MIQP cost time" entry has been removed from the computation-time table (Table X); the table now reports the CSO solving time of the GPU-parallel CEM over all candidates, consistent with Table V.

Q2: The SCM assumptions and error bound are too weak

The surrogate contact model relies on fixed object-environment contact sets and decoupled multi-contact dynamics. These are strong assumptions, especially during flipping, tipping, rolling, or simultaneous multi-contact interaction. Lemma 3 only gives a boundedness argument; it does not establish accuracy or useful dependence on contact geometry, active-set changes, force magnitude, or the off-diagonal blocks of the Delassus matrix.

The paper should state the regularity assumptions required for Lipschitz continuity of the LCP solution and provide empirical SCM error validation against a higher-fidelity contact solver.

Answer: Thank you for this important comment. We agree that SCM relies on strong assumptions and should include a more thorough analysis of modeling errors. Our motivation is that high-fidelity contact dynamics and complete contact detection are both nonsmooth and computationally expensive for optimization. Moreover, high-fidelity contact dynamics are highly geometry-dependent and involve nonsmooth contact transitions, which can introduce numerical instability during optimization. Therefore, the role of SCM is only to predict the correct object motion trend rather than provide high-fidelity contact simulation. In practical deployment, the influence of SCM approximation errors on SCSP is further alleviated by the high-frequency receding-horizon updates of the framework.

We need to correct our previous analysis. The error-bound lemma can become very loose when the Delassus matrix is rank deficient, because its smallest eigenvalue can be dominated by the artificial regularization term. In addition, a global Lipschitz-continuity statement for the LCP solution is difficult to justify when the contact set switches. We therefore removed the erroneous lemma and proof and replaced this theoretical claim with empirical contact-model evaluation.

We evaluate one-step contact-model predictions over sliding, sticking, separation, and contact onset, with samples additionally grouped by contact count $n_c \in \{0, 1, 2, \geq 3\}$ and contact geometry (point, line, surface, and complex). The benchmark uses a sphere, a cube, a cylinder, and seven complex objects, with 200 trajectories per object. One SCM step at $h = 0.02\,\mathrm{s}$ and ten MuJoCo (MJWarp) or IsaacGym (PhysX) steps at $h = 0.002\,\mathrm{s}$ cover the same prediction horizon from identical initial states and physical parameters. SCM uses Eq. (27) with $K = 3$, $\epsilon = 10^{-8}$, and $T = 10^{-4}\,\mathrm{N\,s}$. Each simulator's trajectories determine its contact-regime and contact-count labels; for IsaacGym, SCM uses PhysX-native initial collision manifolds. Table IV (Table IV of the revised manuscript) reports absolute endpoint position and orientation errors (mean $\pm$ std) in millimeters and degrees, respectively, with subgroup sizes.

The results show that SCM maintains millimeter-scale mean position errors over the $20\,\mathrm{ms}$ horizon. The overall position and orientation errors are $0.847 \pm 0.830\,\mathrm{mm}$ and $0.640 \pm 0.914°$ against MJWarp, and $1.220 \pm 0.980\,\mathrm{mm}$ and $0.993 \pm 1.314°$ against IsaacGym. We further generate ideal contact-wrench trajectories with each contact model and execute them in the manipulation evaluation. The ideal success rates are all high (0.99–1.00 over 100 rotation and 100 flipping trials per model), showing that the generated wrench trajectories are generally feasible; higher-fidelity contact models do not automatically produce better deployed performance. In the direct-force evaluation reported in the manuscript's Table V, we compare SCM with the native contact model of each engine, M1 (MuJoCo's QP-based model) and M2 (IsaacGym's PhysX model), used for contact-dynamics prediction in CSO, with collision checking performed in IsaacGym for all models. SCM achieves a manipulation success rate of 0.85 versus 0.75 for M1 and 0.77 for M2, i.e., the best result among the three models. We therefore select SCM for CSO.

Q3: The ranking strategy appears internally inconsistent

The paper defines

$$\rho = \frac{\ell_{\text{near}} - \hat{\ell}_{\text{min}}}{\max(\hat{\ell}_{\text{max}} - \hat{\ell}_{\text{min}}, \epsilon)}. \tag{2}$$

With this definition, smaller ρ means that p_{near} is closer to the best candidate, and larger ρ means it is worse. However, Eq. (39) sets $\kappa_t = \mathbf{1}(\rho_t > \bar{\rho})$ and states that $\kappa_t = 1$ means p_{near} is sufficiently good. This reverses the intended meaning. Eq. (40) is also difficult to reconcile with Algorithm 2 and the described lifting/placing phases.

This section should be rewritten as a clear finite-state or hysteresis rule. As written, the trigger logic is not mathematically consistent.

Answer: Thank you for pointing out this inconsistency. We agree that the original definition of the ranking score was unclear and could lead to a reversed interpretation. In the revised manuscript, ρ_t is explicitly defined as the normalized ranking score of p_{near} based on its cost reduction $\Delta\ell_t^{\text{near}}$ and the range of cost reductions $[\Delta\ell_t^{\min}, \Delta\ell_t^{\max}]$ among all candidate contact locations:

$$\rho_t = \Pi_{[0,1]}\left(\frac{\Delta\ell_t^{\text{near}} - \Delta\ell_t^{\min}}{\max(\Delta\ell_t^{\max} - \Delta\ell_t^{\min}, \epsilon)}\right) \tag{3}$$

It represents the relative ranking of the improvement achieved by p_{near} compared with other candidates. Therefore, a smaller ρ_t indicates that p_{near} provides less improvement and has a worse ranking, while a larger ρ_t indicates that p_{near} is more competitive.

The revised score ranks candidates by the CSO pose objective J_t^i. Its ordering is opposite to the earlier normalized absolute-cost score. Specifically, since the cost reduction is defined as $\Delta\ell_t^i = \ell_{\text{pose}}(\boldsymbol{x}_t) - J_t^i$, with $J_t^i = \ell_{\text{pose}}(\hat{\boldsymbol{x}}_t^{+,i})$, we obtain

$$\rho_t = \Pi_{[0,1]}\left(\frac{J_t^{\max} - J_t^{\text{near}}}{\max(J_t^{\max} - J_t^{\min}, \epsilon)}\right). \tag{4}$$

Here $J_t^{\min}$ and $J_t^{\max}$ are the minimum and maximum candidate pose costs. Thus a lower candidate objective yields a higher ranking score. To avoid this confusion, we have rewritten the Ranking Strategy subsection in Section IV-B and reformulated ρ_t directly based on the cost reduction $\Delta\ell$. The revised manuscript also replaces the earlier counter-based switching rule with EMA smoothing of the binary trigger, $\bar{\kappa}_{t+1} = (1-\beta)\bar{\kappa}_t + \beta\kappa_t$, and thresholds the filtered trigger to obtain the binary phase variable $\gamma_{t+1} = \mathbf{1}(\bar{\kappa}_{t+1} \geq 1/2) \in \{0, 1\}$ used by CPO (Eq. (36)): $\gamma_t = 0$ selects the lifting phase toward $\hat{p}_r^*$ and $\gamma_t = 1$ selects the placing phase toward p_{near}, so Eq. (36), Algorithm 2, and the lifting/placing phases now share one explicit rule. To remove the previous smoothing strategy's dependence on contact estimation—in particular, the T_3 counter that counted consecutive contact steps—we use an EMA that is independent of contact-state estimates.

Q4: The CPO potential costs appear incorrectly specified

The lifting cost uses

$$\ell_{\text{obs}} = \log(\|p_{\text{ee}} - p_o\|^2 + \epsilon), \tag{5}$$

inside a distance threshold and minimizes it. This is not a repulsive potential unless the weight is negative, which is not stated. The placing cost says that ℓ_{con} attracts the end-effector to the reference contact point p_{ref}, but the formula uses p_o instead of p_{ref}:

$$\ell_{\text{con}} = \log(\|p_{\text{ee}} - p_o\|^2 + \epsilon). \tag{6}$$

This should be corrected or explained. A natural formulation would explicitly include a term such as $\|p_{\text{ee}} - p_{\text{ref}}\|^2$.

Answer: Thank you for identifying these inconsistencies. We have revised Section IV-C (highlighted in red). Specifically, the sign of the object repulsive potential ℓ_{obs} has been corrected to ensure a repulsive behavior, and the contact attraction term ℓ_{con} has been modified to attract the end-effector toward an inward offset from the reference contact point $\boldsymbol{p}_{\text{ref}}$ along its surface normal, with $\delta = 0.02\,\mathrm{m}$. We also removed the previous attraction term toward the object geometric center $\boldsymbol{p}_o$. That term was intended to maintain sufficient contact under contact-estimation errors, but it could oppose the attraction

TABLE IV
ABSOLUTE PREDICTION ERRORS OF SCM (MEAN ± STD).

Comparison	Metric	Contact regime				Contact count				Contact geometry			
		Sliding $n = 2000$	Sticking $n = 2000$	Separation $n = 2000$	Onset $n = 2000$	$n_c = 0$ $n = 1788$	$n_c = 1$ $n = 387$	$n_c = 2$ $n = 488$	$n_c \geq 3$ $n = 5337$	Point $n = 946$	Line $n = 439$	Surface $n = 1015$	Complex $n = 5600$
SCM–MJWarp	Position (mm)	0.65 ± 0.81	0.69 ± 0.94	1.63 ± 0.42	0.42 ± 0.42	1.77 ± 0.00133	0.22 ± 0.42	0.46 ± 0.53	0.62 ± 0.79	0.70 ± 0.73	0.72 ± 0.72	1.34 ± 1.07	0.79 ± 0.77
	Orientation (°)	1.08 ± 1.12	0.80 ± 1.03	0.11 ± 0.40	0.58 ± 0.61	0.00136 ± 0.00262	0.40 ± 0.80	0.90 ± 0.95	0.85 ± 0.97	0.40 ± 0.81	1.18 ± 0.85	1.13 ± 1.41	0.55 ± 0.76
		$n = 483$	$n = 118$	$n = 2398$	$n = 419$	$n = 3384$	$n = 731$	$n = 1273$	$n = 2612$	$n = 946$	$n = 439$	$n = 1015$	$n = 5600$
SCM–IsaacGym	Position (mm)	1.18 ± 1.31	0.27 ± 0.42	1.46 ± 0.56	0.43 ± 0.47	1.35 ± 0.76	0.82 ± 0.83	0.69 ± 0.72	1.42 ± 1.22	0.97 ± 0.88	1.08 ± 0.94	1.89 ± 1.15	1.15 ± 0.91
	Orientation (°)	0.96 ± 1.00	0.18 ± 0.34	0.23 ± 0.46	0.59 ± 0.54	0.48 ± 0.95	0.66 ± 0.77	1.09 ± 1.17	1.70 ± 1.55	0.13 ± 0.36	1.44 ± 1.72	1.25 ± 1.30	1.06 ± 1.32

toward $\boldsymbol{p}_{\text{ref}}$ and create a force balance that traps the optimizer in a local minimum. The revised placing cost is:

$$\ell_{\text{con}} = \left\| \boldsymbol{p}_{\text{ee}} - (\boldsymbol{p}_{\text{ref}} - \delta \boldsymbol{n}_{\text{ref}}) \right\|^2 . \tag{7}$$

This removes the conflicting center-attraction term and yielded good empirical performance in the revised experiments.

Q5: The result analysis does not isolate the proposed contributions

The experiments mainly show that the full SCSP system outperforms several baselines in aggregate success rate. They do not sufficiently explain which proposed mechanism causes which improvement. The causal link between the stated contributions and the results is therefore incomplete.

In particular, the paper should report evidence for each component:

- CSO: quality of selected contact priors, contact-location maps, correlation between CSO cost and execution success, and comparison against random, nearest, heuristic, and multi-start contact selection.
- SCM: prediction error against a higher-fidelity contact solver, separated by contact count, contact mode, object geometry, and flipping/rotation phase.
- Ranking strategy: switch frequency, time spent tracking $\hat{p}_r^*$ versus p_{near}, failure cases caused by switching, and sensitivity to $\bar{\rho}, T_1, T_2, T_3$.
- CPO objective: ablation of lift, place, object-center attraction, and p_{ref} tracking terms.
- Robustness: failure categories separating perception error, contact estimation error, slippage, control saturation, and poor prior selection.

Without these analyses, claims about robustness, diversity, and the specific benefit of each module remain qualitative.

Answer: For CSO, we directly apply the optimized contact wrench at the selected contact and roll out the manipulation. The direct-force results in Table V achieve nearly 100% ideal success for SCM, which supports the quality of the selected contact priors. Figure 9 visualizes the contact maps for the elephant and bunny flipping tasks, including the CSO-optimal contact p_r^*, the nearest contact p_{near}, and the physical contact observed during execution. In Section V-B, "Functional Analysis of SCSP Components," A1 follows the optimal CSO contact, A2 always uses the nearest contact, and A3 uses a coarse heuristic contact model; these results isolate optimal, nearest, and heuristic selection. We did not add separate random and multi-start baselines: CSO already evaluates the candidate set in parallel, which provides the multi-start computation pattern, while purely random contacts are too unreliable to yield a meaningful executable comparison.

For SCM, Table IV reports absolute endpoint position and orientation errors over a $20\,\text{ms}$ prediction horizon, grouped by contact regime (sliding, sticking, separation, and onset), contact count $n_c \in \{0, 1, 2, \geq 3\}$, and contact geometry (point, line, surface, and complex). The benchmark uses a sphere, a cube, a cylinder, and seven complex objects, with 200 trajectories per object, and the native contact solvers of MuJoCo (MJWarp) and IsaacGym (PhysX) as references. The direct-force rollout results in Table V further test whether the predicted contact wrenches are executable; the high ideal success rates across models show that the trajectories are generally feasible, while the deployed results show that higher model fidelity alone does not guarantee better task performance.

For the ranking strategy, Table VI reports the CSO-optimal-point switching frequency and physical contact-switching frequency. Table II compares execution time for SCSP with A1 and A2, showing the efficiency difference between ranking-based execution, strict optimal-contact tracking, and nearest-contact execution. Table VI also reports the normalized advantage of an individual ranking decision, computed from a time-discounted success reward, thereby measuring its effect on execution efficiency and outcome. Frequent contact switching in A1 makes effective contact maneuvers difficult to complete, leading to timeouts. Tables VIII and IX report real-world contributing factors and execution errors, including control saturation and slippage, and the fraction of affected trials that ultimately succeed. Finally, Fig. 10 evaluates sensitivity to the acceptance threshold $\bar{\rho}$ and EMA coefficient β. The earlier T_1, T_2, T_3 counter-based smoothing is replaced by contact-estimation-independent EMA smoothing.

For CPO, A6 removes the lift-and-place cost and directly solves an obstacle-avoidance trajectory-optimization problem, while A7 solves the CIMPC formulation without CSO guidance using MPPI (see our reply to Q6). In the current formulation, the contact target is an inward-offset (penetrating) point $\boldsymbol{p}_{\text{ref}} - \delta \boldsymbol{n}_{\text{ref}}$, rather than an attraction term toward the object center. In the real-world ablation A8, we replace this inward-offset target with the estimated surface point $\boldsymbol{p}_{\text{ref}}$; this surface-attraction design often prevents the robot from descending to establish contact. Table VIII reports the resulting contact-estimation-related errors together with the success and recovery outcomes. Tables VIII and IX report the affected-trial counts and recovery rates for perception error, contact-

estimation error, control saturation, and poor prior selection, and for slippage and contact switching, respectively. An event does not necessarily imply final failure because SCSP replanning can sometimes correct it; the recovery rate therefore counts the trials that still succeed after the event. Together with the CSO, SCM, ranking, and hardware analyses above, these experiments quantify the proposed components within the tested tasks.

Q6: Baseline coverage and comparison protocols need strengthening

The Sampling comparison is not fully controlled because the paper replaces the original contact model with f_{cf}. This may create a new baseline rather than a direct implementation of the cited method. The DyWA comparison also needs careful framing: DyWA is described in Table I as lacking generalization, although its stated purpose is generalizable nonprehensile manipulation. The paper should define exactly what "generalization" means.

The baseline set is also incomplete relative to the scope of the claims. The following comparisons or discussions are particularly relevant:

- STOCS / STOCS-3D: this is the closest prior work in spirit, since it directly addresses simultaneous trajectory optimization and contact selection with high-fidelity geometry. Even if it is not real-time, it should be used as an offline planning-quality reference or excluded with a clear justification.
- ContactSDF: this should be discussed for the paper's multi-contact, differentiable contact-model, and dexterous-manipulation claims. If a direct comparison is not feasible, the paper should explain the domain difference and narrow the multi-contact claims.
- Generic sampling-based optimizers: the paper should compare against MPPI, CEM, CMA-ES, or random shooting under the same dynamics model, cost, action parameterization, horizon, and wall-clock budget. Otherwise it is unclear whether SCSP's improvement comes from its contact-selection structure or simply from spending computation on sampled contact/control candidates.

A useful protocol would include both contact-location sampling baselines, e.g., CEM/CMA-ES over p_r and λ_r, and control-sequence sampling baselines, e.g., MPPI/CMA-ES over end-effector waypoints or joint commands without CSO. These should be run with the same model and compute budget as SCSP.

Answer: We thank the reviewer for the valuable suggestions. We have revised the baseline implementation and clarified the comparison protocols in the revised manuscript.

We now fully reproduce the released Sampling implementation, including its C3 contact model, the MIQP solved with Gurobi, and its original contact-selection strategy. The updated comparison results are reported in Table II.

Our use of "generalization" refers to cross-task generalization, whereas DyWA's reported setting concerns cross-object generalization. Table I has been revised accordingly: the term "generalization" is no longer used, and the new column "Task Diversity" refers explicitly to cross-task coverage, under which DyWA and CRISP are marked with a cross, while STOCS-3D and Sampling are marked "Not Validated". We also now credit STOCS-3D and DyWA with active contact selection, and the previous "No Obj.-Specific Analytic Geom." column is renamed "Irregular Geometry".

We additionally include STOCS-3D and ContactSDF in the comparison. STOCS-3D is a CITO method that cannot be evaluated online in our setting and does not switch contact points during execution. Given the initial and target object poses, we compute one contact trajectory offline with STOCS-3D and execute it through OSC. ContactSDF provides a differentiable contact model but does not perform explicit contact selection. The resulting comparison is reported in Table II.

We also evaluate direct sampling-based alternatives. A5 uses CMA-ES to optimize the object-surface contact point and contact force directly, with the SDF distance-to-surface constraint $d_{\mathrm{SDF}} = 0$. A7 solves the CIMPC formulation without CSO guidance, using MPPI: the ranking strategy and the lifting phase are removed, and the objective keeps the attraction toward p_{near}, control regularization $w_u\|\boldsymbol{u}_k\|^2$, and the terminal object cost $w_{\mathrm{cso}}\ell_{\mathrm{pose}}$, with the same cost weights, horizon, contact model f_{cf}, and control bounds as CPO ($N = 5$, 128 samples, one iteration, noise $\sigma = 0.003\,\mathrm{m}$, temperature 0.005). No CSO output is used. The results are included in Table II. A5 remains below SCSP in both tasks (0.89 versus 0.95 for rotation and 0.60 versus 0.81 for flipping), indicating that more refined continuous optimization over the object surface does not by itself improve task success. Although MPPI explores locally more effectively than gradient-based methods such as CF-MPC, it still has difficulty discovering large-range contact switching, so its success rate remains limited. A7 shares the horizon, dynamics model, control bounds, and shared cost weights with CPO. Its lower success rate supports the benefit of contact guidance under the tested settings; the comparison also changes the optimizer from SQP-RTI to MPPI.

Q7: Multi-contact and generalization claims are overstated

The notation allows multiple robot contact points, and the SCM includes multiple environment contacts. However, the main nonprehensile manipulation experiments use point-contact end-effectors, and the SCM explicitly neglects coupling between environment contacts. The later soccer, grasping, and long-horizon experiments are useful case studies, but they do not establish a general simultaneous multi-contact planning capability.

The claim that SCSP can be easily extended to multi-contact tasks should be weakened. The paper should discuss scalability with respect to the number of robot contacts, since selecting n_r contacts from n_s candidates can scale combinatorially, e.g., $\binom{n_s}{n_r}$ in the unordered case.

Answer: Thank you for pointing out this important limitation. We agree that extending SCSP to general multi-contact tasks faces scalability challenges. In Section VI-A(3), the grasping case study does not search over all contact combinations. The CSO search is restricted to the top-5 GWB-based region pairs

plus the points nearest to each fingertip, and the inner loop minimizes a wrench residual over a finite set of sampled disturbance wrenches (Eq. (47)); a small residual on these samples does not certify six-dimensional force closure. Within this restricted search, SCSP selects finger contacts online without manually specifying contact locations, which indicates the potential, rather than a general capability, of extending contact selection and planning to multi-contact manipulation.

However, we acknowledge that more complex multi-contact tasks involve not only combinatorial growth but also strong geometric coupling among contact locations, which remain challenging for the current SCSP framework. The demonstrated precision grasping case involves a limited number of contacts with relatively weak coupling among fingers, while more complex tasks such as power grasping remain challenging for autonomous planning. We have therefore weakened our claims in the revised manuscript, stating that SCSP provides a potential framework for multi-contact manipulation rather than a general solution. Section VII now states explicitly that selecting n_{r} robot contacts from n_s candidates grows combinatorially, e.g., $\binom{n_s}{n_{\mathrm{r}}}$ in the unordered case, and that reducing the CSO search space, as done with the GWB-based regions, is one way to mitigate this growth. This limitation and the scalability of multi-contact planning are also discussed as future work in Section VII.

Q8: Hardware evidence is encouraging but statistically limited

The real-world experiments use only five trials per object/surface condition. This is too small to support strong robustness claims. The paper should either increase the number of trials or report confidence intervals and failure categories. It should also report how often failures come from perception, contact estimation, slippage, control saturation, or poor contact prior selection.

Answer: Thank you for pointing out this limitation. Across the smooth and rough surface conditions, we test five task types—rotation of the bunny, rotation of the duck, flipping of the bunny, flipping of the hand, and flipping of the Xbox controller—with 30 trials for each task/surface condition, for 300 real-world trials per method in total. Table VII reports the success rates and two-sided Wilson 95% confidence intervals.

We also classify unsuccessful trials by their terminal outcomes, including timeout and falling off the designated workspace/table. For perception error, contact-estimation error, slippage, control saturation, and poor contact-prior selection, an individual event does not necessarily cause final failure because SCSP replanning can sometimes recover. Therefore, Tables VIII and IX report both the number of affected trials and the recovery rate, namely the fraction of trials that still succeed after the corresponding event.

Q9: Technical Corrections

- The friction constraints in Eq. (32) should include both signs of each tangential component, e.g.,

$$|\hat{\lambda}^{t1}_{r,i}| \le \mu_i \hat{\lambda}^{n}_{r,i}, \qquad |\hat{\lambda}^{t2}_{r,i}| \le \mu_i \hat{\lambda}^{n}_{r,i}. \tag{8}$$

- SCM is called "smooth," but the max operator is piecewise linear and nonsmooth at the switching surface.
- Eq. (35) should be rechecked for units and consistency between force and impulse notation.
- The text refers to the optimal contact location as $\hat{\lambda}^*_r$ in one place; this should be $\hat{p}^*_r$.
- The quaternion error in Table II is labeled as radians, but the reported metric appears to be $1-(\psi^T\psi_g)^2$, not an angular error in radians.
- Lemma 4 is not a proof of non-conflicting objectives. It should be removed, weakened, or replaced with a precise first-order condition.

Answer: We thank the reviewer for the careful checking of the technical details. We have revised Eq. (32) and Eq. (35) (now Eq. (29) and Eq. (28) in the revised manuscript) to ensure the correctness of the friction constraints and the consistency of force/impulse notation. We have also corrected the notation error after Eq.(32) by replacing $\hat{\boldsymbol{\lambda}}^*_r$ with $\hat{\boldsymbol{p}}^*_r$, removed the incorrect radian unit annotation in Table II, and clarified that the quaternion error metric is a dimensionless quantity.

We agree that the $\max$ operator is nonsmooth at the switching surface. The current manuscript replaces it with the componentwise smooth approximation $\mathrm{softplus}_{T/h}(x) = (T/h)\log(1+e^{hx/T})$ and refines the diagonal estimate with $K = 3$ unrolled Gauss–Seidel sweeps over the contacts (Eq. (27)), using the smooth projection $\Pi^{T/h}_{\mathcal{K}_i}$ (softplus on the normal component and a smooth clip on the tangential components) instead of hard clipping. Lemma 2 and the text following Eq. (27) are restated accordingly: for a fixed contact set, the SCM map is smooth and closed-form in $\boldsymbol{\lambda}_{\mathrm{r}}$, and nonsmoothness arises only when the contact set changes.

We agree that the previous argument was not rigorous enough to support the claimed result. Lemma 4 and its proof have therefore been deleted from the current manuscript.

Reviewer 3

Q1: Contradictory ranking-strategy definitions

The ranking strategy contains contradictory definitions as printed. In (38), ρ is normalized so $\rho \sim 0$ means p_{near} is close to the best candidate, and (39) sets $\kappa = 1$ when $\rho > 0.75$, i.e., $\kappa = 1$ marks p_{near} as a poor candidate. Under this reading, (40), Fig. 6, and Algorithm 2 are mutually consistent (infeasible $\rightarrow$ $\gamma = 0$ $\rightarrow$ lift toward the CSO optimum). But the sentence directly below (39) states the opposite, that $\kappa = 1$ indicates p_{near} is "sufficiently good," and the sentence introducing the trigger defines "a trigger variable γ" and then writes the formula for κ. In addition, both branches of (40) gate on the same counter $t_{(\kappa=1)}$, defined after Algorithm 2 as the total timesteps satisfying $\kappa = 1$; I suspect the first branch should use $t_{(\kappa=0)}$, and it is not stated whether the counters are cumulative or consecutive, or when they reset. As written, the placing branch triggers only after T_1 accumulated infeasible steps, which does not match the described behavior. Please correct the prose and (40) from the implementation and define the counter semantics; at present the switching logic cannot be implemented unambiguously.

Answer: Thank you for the careful observation. We agree that the original formulation contained inconsistencies between the definition of ρ, the interpretation of κ, and the switching logic. The ambiguity mainly originated from a derivation error when simplifying the original formulation.

In the revised manuscript, we have rewritten Section IV-B and reformulated ρ_t with a clearer definition. Specifically, for each candidate contact location $p_i \in \mathcal{I}_{\text{avail}}$, the optimal contact force $\boldsymbol{\lambda}_i^*$ is solved to obtain the corresponding optimal cost reduction $\Delta\ell_t^i = \ell_{\text{pose}}(\boldsymbol{x}_t) - J_t^i$, where $J_t^i = \ell_{\text{pose}}(\hat{\boldsymbol{x}}_t^{+,i})$ is the CSO pose objective. The ranking score ρ_t then measures the relative ranking of the cost reduction obtained by p_{near} among all candidate contact locations. Therefore, ρ_t is reformulated as follows:

$$\rho_t = \Pi_{[0,1]}\left(\frac{\Delta\ell_t^{\text{near}} - \Delta\ell_t^{\min}}{\max(\Delta\ell_t^{\max} - \Delta\ell_t^{\min}, \epsilon)}\right). \tag{9}$$

With this definition, a larger ρ_t indicates that p_{near} provides greater improvement and has a better ranking. Accordingly, $\kappa_t = \mathbf{1}(\rho_t \geq \bar{\rho})$ consistently indicates whether p_{near} is sufficiently good. The previous derivation error was introduced during a further simplification of the formulation, as shown in (4).

The current manuscript replaces the counter-based rule with exponential moving-average smoothing of the binary trigger, $\bar{\kappa}_{t+1} = (1-\beta)\bar{\kappa}_t + \beta\kappa_t$ with $\bar{\kappa}_0 = 0$, followed by a fixed threshold, $\gamma_{t+1} = \mathbf{1}(\bar{\kappa}_{t+1} \geq 1/2)$ (Eq. (36)). The phase variable $\gamma_t \in \{0, 1\}$ is therefore binary: $\gamma_t = 0$ activates the lifting cost toward $\hat{p}_r^*$ and $\gamma_t = 1$ activates the placing cost toward p_{near}, which is the explicit switching rule used in Algorithm 2 and Fig. 6. There are no counters, and the only state is the filtered trigger $\bar{\kappa}_t$, which is updated at every planning step and never reset. The earlier smoothing still depended on contact information, in particular the T_3 counter of consecutive contact steps, which is not robust to contact-estimation errors in the real-world experiments. The EMA does not require any contact-related input and reduces the tuning burden by replacing the multiple counter thresholds with the single smoothing coefficient β. Its feasibility is evaluated in the real-world experiments in Section V-C and in the sensitivity study in Fig. 10.

Q2: Optimization claims

CSO performs a global search over the sampled candidate set, not over the continuous surface, so "globally optimal" (Fig. 2 caption) and especially "asymptotically optimal" (after (32), where the symbol should also be $\hat{p}_r$, not $\hat{\lambda}_r$) are overclaims; please reword or prove. The paper calls the problem an MIQP with sub-millisecond QP subproblems, but the implementation uses SNOPT, an SQP solver for NLPs, and with the quaternion cost the subproblem is not a QP. Algorithm 1 is titled "Sampling and Exchange Method" and Table V lists an "Exchange cost time," but no exchange step is defined anywhere. Since 70×0.23 ms is about the reported 16 ms CSO time, I assume the candidates are simply enumerated; if so, just say so.

Answer: Thank you for pointing out these inconsistencies.

We agree that "asymptotically optimal" was incorrect. CSO searches only the finite sampled candidate set, so the selected contact is optimal among the sampled candidates rather than asymptotically optimal on the continuous object surface. We have changed the wording accordingly and corrected the variable error in Eq. (32) from $\hat{\boldsymbol{\lambda}}_r^*$ to $\hat{\boldsymbol{p}}_r^*$.

The MIQP/QP characterization was also inaccurate. We first need to correct an omission in the previous description of SCM. The revised implementation uses $K = 3$ unrolled Gauss–Seidel sweeps over the environment contacts in a fixed order: starting from $\hat{\boldsymbol{\lambda}}_{\text{env}}^{(0)} = \mathbf{0}$ and the free-motion velocity $\boldsymbol{v}^{(1,0)} = \boldsymbol{v}_{\text{free}}$, sweep $k = 1, \ldots, K$ computes for $i = 1, \ldots, n_{\text{env}}$ the update $\hat{\boldsymbol{\lambda}}_{\text{env},i}^{(k)} = \Pi_{\mathcal{K}_i}^{T/h}\big(\hat{\boldsymbol{\lambda}}_{\text{env},i}^{(k-1)} - \boldsymbol{D}^{-1}(\tilde{\boldsymbol{J}}_{\text{env},i}\boldsymbol{v}^{(k,i-1)}/h + \boldsymbol{d}_{\text{env},i}/h^2 + \epsilon\hat{\boldsymbol{\lambda}}_{\text{env},i}^{(k-1)})\big)$ and $\boldsymbol{v}^{(k,i)} = \boldsymbol{v}^{(k,i-1)} + h\boldsymbol{M}_o^{-1}\tilde{\boldsymbol{J}}_{\text{env},i}^{\top}(\hat{\boldsymbol{\lambda}}_{\text{env},i}^{(k)} - \hat{\boldsymbol{\lambda}}_{\text{env},i}^{(k-1)})$, with $\boldsymbol{v}^{(k+1,0)} = \boldsymbol{v}^{(k,n_{\text{env}})}$, $\boldsymbol{d}_{\text{env},i} = \max(\tilde{\boldsymbol{\phi}}_{\text{env},i}, \mathbf{0})$, and output $\hat{\boldsymbol{\lambda}}_{\text{env}} = \hat{\boldsymbol{\lambda}}_{\text{env}}^{(K)}$, so that each contact force is evaluated from the object velocity already updated by the preceding contacts and sweeps, which was not stated in the previous manuscript. These sweeps are now given explicitly in Eq. (27). Second, with these sweeps the force map is not piecewise linear: the componentwise $\max$ of the previous version is replaced by the smooth projection $\Pi_{\mathcal{K}_i}^{T/h}$, which applies $\text{softplus}_{T/h}$ to the normal component and a smooth clip to the tangential components, so for a fixed contact set the map $\boldsymbol{\lambda}_{\text{r}} \mapsto \hat{\boldsymbol{\lambda}}_{\text{env}}$ is smooth and closed-form (Lemma 2 and the text following Eq. (27)); nonsmoothness arises only when the contact set changes. The nonlinear softplus, the sequential sweeps, and the quaternion cost together prevent a QP formulation. Nevertheless, the resulting optimization remains lightweight because the model is smooth and closed-form for a fixed contact set and the candidate set is finite.

The previous implementation solved the continuous subproblem serially with SNOPT, which was already sufficiently fast for the tested setting. The current implementation replaces this serial solve with GPU-parallel CEM and obtains higher solving efficiency. No binary variables, big-M constraints, or mixed-integer solver are used.

Algorithm 1 describes offline FPS followed by online valid-region selection. We have defined the Exchange step explicitly in Section III-B: it applies the offline-specified rule-based and manual masks online and exchanges the convex approximation $\mathcal{F}$ for the retained valid surface $\mathcal{F}_{\text{v}}$, from which $\mathcal{I}_{\text{avail}}$ is selected. The "Exchange cost time" entry in the computation-time table reports the cost of this masking step; it is not a separate optimization. The candidates are then simply enumerated, as the reviewer surmised.

Q3: Text vs. tables

Sec. V-C states SCSP achieved a 100% success rate in on-ground rotation, but Table IV shows 19/20 (Duck, rough surface: 4/5). The open-loop numbers for the long-horizon tasks are missing from Table VI although the text calls the rates "comparable." Tables IV and VI say "(%)" but list raw counts. "Significantly outperforms" is not accurate for multi-finger grasping (16/20 vs 15/20). Please reconcile all of these. I would also suggest confidence intervals: with n = 200 they

are trivial to add, and with n = 5 per hardware cell the current wording is stronger than the data supports.

Answer: Thank you for the suggestions.

We have corrected all of these inconsistencies. The real-world evaluation now uses 30 trials for each object/surface condition, and the revised success rates and two-sided Wilson 95% confidence intervals are reported in Table VII. The open-loop long-horizon results are included in Table XI, and the wording for multi-finger grasping has been corrected to avoid overstating the comparison.

Q4: Reproducibility

The friction-cone constraints in (32) have no lower bounds on the tangential components as printed (and n_t vs n_d is inconsistent with (9)); please print the exact constraints used. Values for w_{att}, w_{obs}, w_{con}, w_o (is this the same as w_{obj}?), σ, the lift offset d, and solver settings are missing. Please also mark the 4.82 s sampling in Table V as offline so the real-time claim is unambiguous.

Answer: Thank you for pointing out these issues. We have added the concrete settings to the experimental-setting description in Section V. The exact friction constraints are now printed in the CSO formulation, $|\hat{\lambda}^{t1}_{r,i}| \le \mu_i \hat{\lambda}^{n}_{r,i}$ and $|\hat{\lambda}^{t2}_{r,i}| \le \mu_i \hat{\lambda}^{n}_{r,i}$ with $0 \le \hat{\lambda}^{n}_{r,i} \le \hat{\lambda}^{n}_{\max}$. The lifting and reference-contact attraction weights are $w_{\text{att}} = w_{\text{con}} = 20$. The repulsive-field term is disabled in the reported experiments, so w_{obs} and σ are not used; the previous object-center attraction term and its weight w_{obj} have been removed from the CPO cost (see our reply to Reviewer 1, Q4). The single lift offset d is replaced by the Bézier detouring curve with apex height $z_{\text{c}} \ge z_{\text{top}} + 0.025$ m and an attractive waypoint advancing smoothly along the curve. The solver settings are also stated: CSO uses GPU-parallel CEM with 20 iterations over $n_s = 70$ candidates, and CPO uses `acados` SQP-RTI with HPIPM and a tolerance of 10^{-4}, with $N = 5$, $|u_j| \le 0.005$ m, $w_{\text{lp}} = 1$, $w_u = 50$, and $w_{\text{cso}} = 10$. We also mark the 4.82 s sampling cost as offline in the computation-time table.

Q5: Minor comments

In (33), α is introduced as a weighting coefficient but never used. $\tilde{Q}$ appears in (8) and (11) but only Q is defined in (4). In (47), the terminal term is $w_{\text{cso}}\ell_{\text{cso}}$ while the text says V_f as in (7); the same equation mixes t and k as time indices. In (13) and (32) the objective carries a free index k; presumably x_T is meant. (18) is undefined when n_i is parallel to the reference axis a; please state the fallback. Please state the orientation tolerance in (51) in degrees (about 26 deg) and the distribution behind "mass randomly scaled within a factor of 10."

Answer: Thank you for pointing out these issues.

All of these points have been corrected in the revised manuscript:

- The two-stage stable-pose objective in the previous (33), together with its unused coefficient α, has been removed. The CSO objective is $\ell_{\text{pose}}(\boldsymbol{x}_T)$, where $\ell_{\text{pose}} = 500\|\boldsymbol{p}_o - \boldsymbol{p}_{\text{goal}}\|_2^2 + 20[1-(\boldsymbol{\psi}_o^\top \boldsymbol{\psi}_{\text{goal}})^2]$. This shared object objective applies to all objects and both nonprehensile tasks; CPO uses $w_{\text{cso}}\ell_{\text{pose}}$ as its terminal cost.
- $\tilde{\boldsymbol{Q}}$ in (8) and (11) was a notation error; it is now $\boldsymbol{Q}$ as defined in (4).
- The CPO formulation (now Eq. (40)) writes the terminal term as $V_f(\boldsymbol{s}_N) = w_{\text{cso}}\ell_{\text{pose}}(\boldsymbol{s}_N)$ and uses the single index $k = 0, \dots, N-1$ throughout.
- The CSO objective (now Eq. (29)) is evaluated at the terminal state $\boldsymbol{x}_T$; the free index k has been removed.
- Eq. (18) now states the fallback: when the reference axis $\boldsymbol{a}$ is nearly parallel to $\boldsymbol{n}_i$, another coordinate axis is selected as the reference axis.
- The orientation tolerance is stated in degrees: the threshold $1-(\boldsymbol{\psi}_{\text{goal}}^\top \boldsymbol{\psi}_o)^2 \le 0.015$ in the success criterion (now Eq. (42)) corresponds to approximately 14.1°. The mass is sampled with equal probability from the discrete set $\{0.01, 0.1, 1.0, 10.0\}$ kg, while SCM uses the fixed matrix $\boldsymbol{M}_o = \text{diag}(50, 50, 50, 0.05, 0.05, 0.05)$.

Reviewer 5

Q1: Quality

- The authors cover a large span of CI-MPC works from recent literature and appropriately denote inspiration for the proposed surrogate contact model.

- The experimentation is thorough and extensive, each validating the overall efficacy of SCSP across a broad range of tasks.

- The ranking strategy of contact locations, optimized by contact selection optimization (CSO) and sliding window switching strategy as described in Alg. 2, and the various attractive and repulsive potentials, described in Eqns. 41-46, are fairly engineered and require significant heuristic tuning, as briefly described at the bottom of page 11.

- Yet, this ranking strategy is the single most important component of SCSP, smoothing out the volatility of contact selection with some heuristics and yielding superior performance compared to the baseline methods. Since the ranking strategy is such an integral component of SCSP, its ablation and contribution characterization should be elaborated on a good deal more. I am curious how often contact switching, as defined in Eq. (37)-(40) with the grid-swept parameters, is actually triggered. Is there a measure of how helpful it is?

- The authors duly note that the greatly simplified surrogate contact model (SCM) employed for fast CSO limits optimization to point contacts and prevents sequential state prediction due to poor simulation fidelity of complex contact/geometry and deformable objects. This is a large limitation, but broadly endemic to CI-MPC.

- Contribution four claims manipulation "using only vision perception, without the assistance of markers or motion capture systems." I think caveating with the one-time manual MuJoCo calibration is necessary here.

Answer: Thank you for recognizing the importance of the ranking strategy and providing valuable suggestions. We have made the following revisions accordingly.

1) Contribution of ranking strategy:

We now characterize the concrete contribution of the ranking strategy in two complementary ways. First, we evaluate overall task success with ablations. A1 strictly follows the CSO-optimal contact p_r^*, while A2 always selects the nearest contact p_{near}. The SCSP success rate is clearly higher than

both monotonic policies in Table II, showing the benefit of switching between the two contact strategies rather than committing to only one.

Second, Section V-B evaluates the value of an individual ranking decision with matched counterfactual rollouts. We define the terminal reward as $r_T = \mathbb{I}(\text{success})$ and the value of a contact-selection decision b at state s_t as

$$J(s_t, b) = \mathbb{E}_{\tau\sim\pi(\cdot|s_t,b)}\left[\eta_{\text{cf}}^{T}\mathbb{I}(\text{success})\right], \qquad \eta_{\text{cf}} = 0.99, \tag{10}$$

where T counts the actual execution steps from s_t to trial termination and the discount factor is denoted by η_{cf} to avoid confusion with the ranking phase variable γ_t. We sample 200 SCSP rollout trajectories and identify 10 decision points per trajectory where p_r^* differs from p_{near} (2000 decision points in total). We reverse each recorded decision and prevent ranking overrides until the selected contact is established, then resume full SCSP until termination. For example, during a contact-free phase, if the ranking strategy selects the nearby contact p_{near}, the counterfactual forces the robot to detour to p_r^*; during a contact phase, if the ranking strategy decides to break the current contact and approach p_r^*, the counterfactual forces the robot to maintain the current contact.

Let J_{RS} and J_{cf} denote the values of the original and reversed decisions. We quantify the decision effect with the normalized advantage

$$\widetilde{A}(s_t) = \begin{cases} \dfrac{J_{\text{RS}} - J_{\text{cf}}}{\max\{J_{\text{RS}}, J_{\text{cf}}\}}, & \max\{J_{\text{RS}}, J_{\text{cf}}\} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{11}$$

The results are reported in Table VI. We report the CSO-optimal contact-point switching frequency f_r^* and the physical contact-switching frequency f_{switch}, characterizing changes in contact guidance and physical execution, respectively. We also report the selection probabilities of p_r^* and p_{near}; they are relatively balanced, showing that the ranking strategy makes substantial use of both contact strategies. The normalized advantage $\widetilde{A}$ measures the value improvement contributed by each individual ranking decision.

2)Poor simulation fidelity: Similar to most CIMPC methods, SCM introduces modeling errors due to contact model simplification, and the point-contact assumption further limits its applicability to deformable objects. However, our experiments show that SCSP maintains robust performance under perturbed physical properties in simulation and without system identification in real-world experiments. This demonstrates that SCSP tolerates single-step simulation errors, which is further supported by our additional analysis experiments. With high-frequency replanning, such simulation errors do not significantly affect the contact selection and planning decisions of SCSP.

SCSP has limitations in highly dynamic tasks and deformable object manipulation. Further extending SCSP may require a higher-fidelity contact model, as discussed in Sec. VII.

3)Clarification of contribution: Although our object pose estimation does not rely on motion capture, we clarify that a one-time calibration estimates only object scale and positional offsets from kinesthetic teaching. The MuJoCo proxy is used for visualization and coarse kinematic alignment; the rendered object pose comes from FoundationPose and object collision simulation is disabled in the proxy, while SCM predicts contact dynamics.

Q2: Clarity

- The bullet points in the "main contributions" paragraph are mostly rehashed from the preceding paragraph beginning with "In this work,..." Only one of these large text blocks is necessary.

- Broadly, Section I uses adjective-dense and redundant phrasing, and it should get a revision for clarity and concision.

- pg. 6 Sec. III.B: how are semantically unsuitable regions for manipulation determined? The other heuristics for valid region selection are similarly underexplained.

- Fig. 9b1-4 legend items and axes are hard to read. To accommodate this, the Fig. 9a1-4 panels can have reduced footprint.

- The unit of time in all the x-axis timesteps is not defined, please clarify. I'm assuming it is 0.02s, the value of h?

- I would rename Case Study II from Long-Horizon Tasks to Articulated Object Manipulation. The opening sentence does not make much sense, particularly the phrase "exhibits physical inference abilities." More simply put could be: "SCSP leverages active contact selection to perform robust articulated object manipulation." Additionally, neither Sec. VI.B paragraph, the theoretical formulation of CSO, nor Fig. 18 provide evidence that active CSO, compared to the open-loop CSO, aids in avoiding singularities.

- Referring to CSO as a world model throughout Sec. VII is understandable but also a case of concept creep. The comparison between proposed CSO to PointWorld [55] is a big jump over simply higher-fidelity physical simulation.

Answer: Thank you for the constructive suggestions. We have revised Section I by removing redundant descriptions and improving the clarity and conciseness of the presentation. We have also adjusted the layout of the simulation figures. For the semantically unsuitable regions in Sec. III.B, they are introduced as optional task-level constraints to improve the generality of SCSP. In practice, the mask combines manual annotation with rule-based generation. Both the manual and the rule-based masks are specified offline together with the object model; no online segmentation or human annotation is performed during manipulation (Section III-B). Manual annotation removes semantically unsuitable regions: in the articulated-object tasks, the object is segmented offline into parts such as handles, and non-manipulable parts are excluded; in the soccer task, points close to the ground ($z_i < 0.02$ m) are masked, and each foot is activated only when the ball lies in its half-space, with workspace constraints imposed on each foot. Rules mask contacts that are inadmissible in the current state: the drawer interior and the microwave-door interior are masked before the drawer or door is pulled open, and for the current top-down end-effector orientation we coarsely sample its Cartesian workspace and mask contact points outside it, since reaching beyond this region brings the arm near singularities and requires large changes in end-effector orientation.

When the handle contact leaves this workspace, SCSP masks it and switches to the drawer inner wall or microwave door (Section VI-B). We clarify that these heuristics are optional components chosen according to the task.

We agree that the original title "Long-Horizon Tasks" is not sufficiently accurate. The presented experiments demonstrate that SCSP's active contact selection and contact switching strategy can be extended beyond conventional push and flip tasks to improve robustness in daily-life articulated object manipulation scenarios. This further suggests the potential of SCSP as a general contact strategy generator for providing diverse manipulation experiences for future learning-based frameworks. Therefore, we have renamed this case study as "Articulated Object Manipulation" and revised the opening statement accordingly. We have also clarified the role of the online feasible-set masking in handling robot feasibility issues. For the current top-down end-effector orientation, contacts outside the sampled Cartesian workspace are masked; SCSP switches contacts when the handle contact leaves this workspace. We have added this clarification in Sec. VI.B (the paragraph accompanying Fig. 19 and Table XI), and removed the overstatement that CSO explicitly avoids singularities; the revised text describes online feasible-set evaluation and workspace masking. Specifically, the revised Fig. 19 and its caption now show that SCSP switches contacts when the current contact leaves the predefined workspace (e.g., from the handle to the drawer inner wall or microwave door). The accompanying paragraph in Sec. VI.B reports that SCSP succeeds in 20/20 trials versus 15/20 for the open-loop baseline (Table XI), that the offline contact sequences fail under slippage or near-singular configurations, and that, SCSP evaluates the feasible contact set online and masks the handle contact once it leaves the workspace. The experimental description in Sec. VI.B also states the coarse Cartesian workspace region and the masking of the drawer and microwave-door interiors that drive these contact switches.

Finally, we agree that the term "world model" in Sec. VII is overly broad. Our original intention was to indicate that SCM could potentially be replaced by a higher-fidelity contact model to extend SCSP to more complex scenarios. Since "world model" has a broader meaning in recent literature, we have removed the term from Sec. VII. The revised discussion refers only to higher-fidelity contact models such as PointWorld as a potential direction for sequential, high-fidelity state prediction, and explicitly notes that their interfaces and task domains differ from SCM, so they are not drop-in replacements.

Q3: Significance

- The core algorithmic simplification presented in the method, namely diagonalization of the Delassus matrix in the SCM for smooth, efficient optimization in CSO, is heavily inspired, as noted by the authors, by Jin's complementarity-free MPC work [27], which somewhat similarly diagonalizes the Delassus matrix as a scaled identity matrix. Where Jin justifies this extreme physical simplification with receding-horizon control, the authors propose a heuristic-driven smoothing over this simplified, open-loop contact selection optimization via the ranking strategy. Both make fairly strong assumptions about rigid bodies and validate the approach empirically.

- The authors provide finite bounds on the error of this simplification in Lemma 3. However, if I am understanding correctly, the defined bounds are very large: $\sigma_1/\lambda_{\text{min}}$ (Eqn. 58), where λ_{min} is the minimal eigenvalue of the Delassus matrix (W_{env}), which under rank-deficient Jacobians (essentially the default case) is determined by the selected, typically very small regularizing term R in Eq. 22, which can be increased but is itself already a physically spurious quantity. I also would appreciate some extrapolation on "standard perturbation theory for LCP." Generally, I think the authors introduce limited theoretical novelty, but that should not detract from the work's strong and broad results, including real-world deployment and practicality (not needing mocap).

Answer:

Thank you for the careful assessment. We are indeed deeply inspired by Jin's complementarity-free simplification: we construct a closed-form contact model so that CSO can evaluate contact candidates efficiently.

We also need to correct an omission in the previous manuscript. The $K = 3$ unrolled Gauss–Seidel sweeps over the environment contacts (Eq. (27)) were not described. Directly constructing D from the diagonal entries of the Delassus matrix is an aggressive simplification and can introduce substantial model bias, which makes the corresponding error bound loose; the sweeps, in which each contact force is evaluated from the object velocity already updated by the preceding contacts and sweeps, correct part of this bias. Nevertheless, as the reviewer observes, the bound can still become very large under rank-deficient Jacobians. SCM can represent separation of retained old contacts, but newly created contacts are absent from the current contact set and inevitably introduce additional prediction bias.

For the same reason, the previous lemma and proof are not useful as general guarantees. The earlier appeal to standard perturbation theory for LCPs was also inappropriate: perturbations near a contact-set boundary can cause large changes, and SCM does not smooth those contact-set transitions. We have therefore removed the error-bound lemma and its proof and no longer present this perturbation argument as a theoretical guarantee.

We instead validate SCM empirically as a feasible contact model for contact selection. Table IV compares one SCM step at $h = 0.02\,\text{s}$ with ten reference-simulator steps at $h = 0.002\,\text{s}$ from identical initial states and physical parameters. It reports absolute endpoint position and orientation errors against MuJoCo (MJWarp) and IsaacGym (PhysX), grouped by contact regime (sliding, sticking, separation, and onset), contact count, and contact geometry. The overall errors are $0.847 \pm 0.830\,\text{mm}$ and $0.640 \pm 0.914^\circ$ against MJWarp, and $1.220 \pm 0.980\,\text{mm}$ and $0.993 \pm 1.314^\circ$ against IsaacGym. We then generate optimal contact-wrench trajectories with each model and test their feasibility in the direct-force evaluation. The contact-guidance results (Table V) show that all models produce largely feasible trajectories, while SCM performs better in deployment than the native engine models (0.85 versus 0.75 for M1 and 0.77 for M2). This demonstrates that model fidelity is not directly

proportional to task success and motivates using SCM for CSO.

Q4: Various typos and minor addenda

Various typos and minor addenda:

- pg. 1: “thus efficiently handle”; “thus efficiently handling”
- pg. 2: linear complementarity system should be defined before the “LCS model” acronym
- I believe [30] and [31] have been conflated into one in the related works and in Table 1.
- pg. 2 “This work inspiring”; “this work is inspiring”, or better “Our approach is inspired by this work,”
- pg. 2 “existing CIMPC formulation struggle”; “existing CIMPC formulations struggle”
- pg. 2 “by fast simulating state transitions” ; “by fast simulation of state transitions”
- Eqn. 6 the iterator i should be k to be consistent with following usage
- pg. 6 “the sampling process is performed offline”; what is the computational cost of sampling?
- pg. 13 Floating “I-MPC” after “quaternion error.” in the second paragraph

Answer: Thank you for the careful reading. We have addressed each item as follows:

- pg. 1 “thus efficiently handle”, pg. 2 “This work inspiring”, “existing CIMPC formulation struggle”, and “by fast simulating state transitions”: Section I has been rewritten for clarity and concision, and these sentences no longer appear in the revised manuscript.
- “LCS model”: the acronym has been removed. The revised Section I refers to linear complementarity problems (LCPs) in full.
- References [30] and [31] (now [31] and [32] in the revised manuscript): the two works are now cited separately in the related-work discussion of Section I (contact-trust-region formulation and motion–contact planning with tracking, respectively), and Table I cites each listed method individually.
- Eq. (6): the iterator i has been changed to k for consistency with the subsequent usage.
- Offline sampling cost: the FPS sampling remains offline, and its measured cost (4.82 ± 3.85 s) is now reported in the computation-time table (Table X) and explicitly marked as offline.
- Floating “I-MPC” on pg. 13: removed.

The revised manuscript also states the simulation time step $h = 0.002\,\mathrm{s}$, the 14.1° orientation tolerance, and the discrete mass distribution used in the experiments.